Structure generation (MatterGen) & DFT labeling
Generation under an energy-above-hull constraint
DFT-computed SSE labels

Binary Altermagnet screening
$|\boldsymbol{M}| \leq 0.005\ \mu_B$ & SSE > 1meV

Passed screening
+ strain sampling

N = 3,851

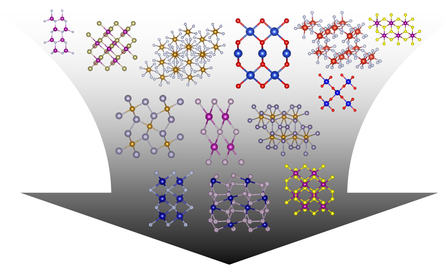
52 DFT-free Descriptors
From structure and composition only
(e.g., local symmetry, bond-length statistics)

XGBoost surrogate for SSE
Predicts SSE from DFT-free descriptors

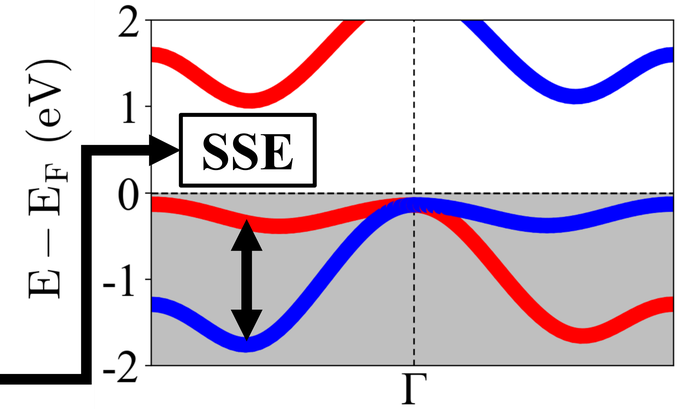
Bayesian inverse design
Search descriptor space
Maximize predicted SSE
Constraint: CN ∈ {4, 6}

DFT validation
Top-5 predicted candidates (CN ∈ {4, 6})
α-NiS identified

MatterGen-generated structures

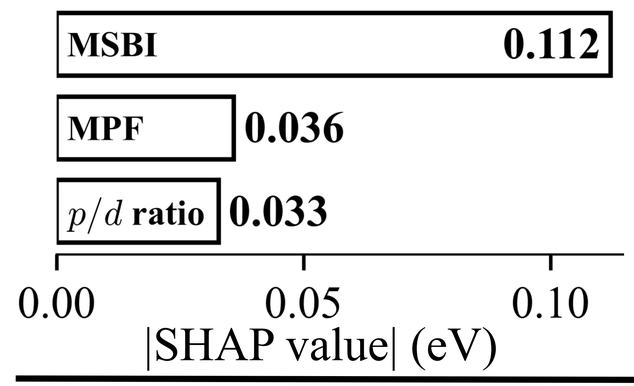


DFT calculation

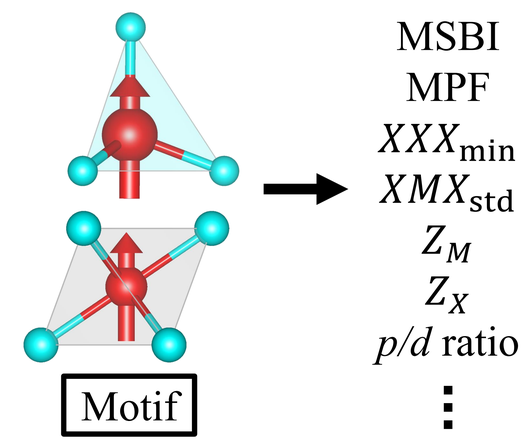


SSE: Maximum spin splitting energy $[E_F - 2\ \text{eV},\ E_F]$

SHAP Interpretation
Top-3 drivers:
MSBI, MPF, *p/d* ratio



Motif-based descriptors



Descriptor-space Bayesian optimization

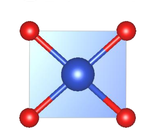
Sampled descriptors

Maximize SSE

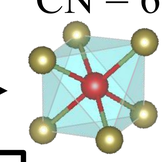
Trained XGBoost surrogate

Top candidates

CN = 4

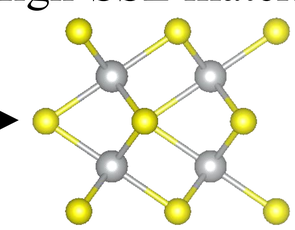


CN = 6



Prototype matching

DFT-validated
High-SSE material



α-NiS
SSE: 0.823 eV